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Project Title:	The Influence of Rotational Cues on Human Tilt and Translation Responses		
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Comments:	In August 2005, the PI was promoted Assistant-Professor in the Dept of Otology and Laryngology at Harvard Medical School.		
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Contact Monitor:		**Contact Phone:**	
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Flight Assignment:	NOTE: PI is now Dan Merfeld (per P. Denkins/JSC), so closing this task (jvp 4/07) NOTE: Funding reinstated to 6/1/2008, per J. Cohen (HQ, 8/2006)		
Key Personnel Changes/Previous PI:	NOTE: Dan Merfeld took over as PI in Spring 2007; for information after FY2007, please see Merfeld's task with the same title. For further information, please contact the Task Book Help Desk, taskbook@nasaprs.com		
COI Name (Institution):	Merfeld, Daniel (Massachusetts Eye and Ear Infirmary)		
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Task Description:

Sensory systems often provide ambiguous information. For example, otolith organs measure gravito-inertial force, the sum of gravitational force and inertial force due to linear acceleration. According to Einstein's Equivalence Principle, no set of linear accelerometers alone can distinguish gravitational force (which changes with head orientation during a head tilt) from inertial force (which changes with linear acceleration of the head). Therefore, the Central Nervous System (CNS) must use other sensory cues to distinguish tilt from translation. For example, the CNS can use rotational cues provided by the semicircular canals and vision. To investigate the influence of rotational cues on the neural processing of tilt and translation in humans, we propose to develop a countermeasure to adapt and/or pre-adapt astronaut canal-otolith interactions to any gravitational environment (Earth, Mars, …) by providing correlated sensory cues between the canals and the otoliths. The countermeasure we propose to develop, often referred to as Artificial Gravity, consists of centrifugation combined with intermittent motion. We will measure behavioral (eye movements such as vestibulo-ocular reflexes) as well as psychophysical responses (orientation estimation) during tilt alone, translation alone or combined tilt-translation motion in either a 1-G or hypergravic environment. Fourier analysis and modeling tools will also be used to investigate the neural processing between rotational and gravito-inertial cues.

This Task addresses the following specific aims (SA) using human subjects:

SA I. Using psychophysics and eye movement measurements, investigate how humans utilize semicircular canal cues to help estimate the relative orientation of gravity (monitored through perceptual tilt responses) as well as the estimate of linear acceleration (monitored through translational VOR responses). This specific aim addresses Critical Question 9.01 since SA 1 will provide baseline data for humans to evaluate if Artificial Gravity combined with intermittent motion is a valid countermeasure (SA 2). We propose a ground study to measure human responses (perceptual tilt and eye movements) during combined canal-otolith stimulation on Earth, in a 1-G environment. Our goal is to provide a perceptual measure of tilt with simultaneous eye movement recording to comprehensively investigate how the human Central Nervous System distinguishes tilt from translation. Subjects will be tested using protocols in which they passively a) tilt, b) translate, or c) tilt & translate simultaneously.

SA II. Using psychophysics and eye movement measurements, investigate how humans utilize semicircular canal cues to help estimate the relative orientation of gravity (monitored through perceptual tilt responses) as well as the estimate of linear acceleration (monitored through translational VOR responses) by measuring adaptive changes induced by exposure to a hyper-G environment. This specific aim addresses both Critical Questions (CQ) 9.01 and 9.04 by evaluating if Artificial Gravity combined with intermittent motion is a valid countermeasure (CQ 9.01) and by measuring adaptive changes induced by exposure to altered-gravity (CQ 9.04). We propose to measure human perception of orientation and eye movement responses during exposure to artificial gravity. To induce adaptation, we propose to utilize intermittent motion during centrifugation: passive roll tilt in a supine orientation will dynamically stimulate both semicircular canals and otolith organs, such that both sensory cues are correlated. (The motion envelope is in an Earth-horizontal plane.)

For Specific Aim #1, we expect the human perceptual responses to be somewhat consistent with those previously measured in monkeys by Angelaki and Colleagues (1999), with the canal cue essential for accurate estimation of gravity and linear acceleration, but we also expect differences in eye responses as shown by our preliminary data with a significant horizontal eye response to roll tilt at high frequencies.

The GIF resolution hypothesis states that the CNS estimates gravity and linear acceleration such that their mathematical difference matches the graviceptor measurement of the GIF. Therefore, for Specific Aim #2, based on the GIF resolution hypothesis, we expect that:

a) the human responses for roll tilt about an earth-vertical axis in a supine orientation with a centripetal 1-G linear acceleration (Expt 2A) to be similar as for roll tilt about an earth-horizontal axis in an upright orientation in a 1-G environment since both dynamic canal and otolith cues are the exact same (except for a static naso-occipital 1-G gravitational acceleration in 2A): such a result would validate artificial gravity combined with intermittent motion as a countermeasure. The responses may not be exactly similar during Trial 1 but we expect the CNS to have adapted by Trial 4 back to a response similar to the response in Expt. 1 (1-G environment).

b) the human responses for roll tilt about an earth-vertical axis in a supine orientation with a centripetal 1.2-G (or 1.4 G) linear acceleration (Expt 2B) to be originally (Trial 1) different from responses for roll tilt about an earth-horizontal axis in an upright orientation in a 1-G environment: for example, we expect a larger vertical VOR response (Trial 1). Then we expect the CNS to adapt to this Hyper-G environment: for example, the vertical VOR response should decrease while the horizontal VOR response increases as the CNS adapts to the hyper-G environment.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Since all proposed experiments are ground studies, the Earth Benefits should be straight forward. Experimental data shoud provide a better insight into how the Central Nervous System separates the graviceptor measurements of gravito-inertial force into estimates of gravity and linear acceleration, or in other words, into estimates of tilt and translation. This central process is essential to a number of everyday task such as visual stabilization or postural control and its understanding is therefore essential to the understanding of these many central processes. In addition, these ground studies should also provide insights on how the Central Nervous System adapts this separation process when humans experience a new gravitational environment. Understanding this process should not only help train and condition astronauts to microgravity or to a return to Earth-gravity after a long stay in microgravity. But it should also help us understand the rehabilitation process in vestibular patients and it should therefore help us design better methods of rehabilitation.

Task Progress:

The PI has completed the third year of this Task. Because of unforeseen technical difficulties, the eccentric rotator necessary to conduct the studies proposed for NASA grant # NNJ04HF79G has not yet been delivered and additional time was dedicated during Year #3 to improve and complete the design of this new device. Since the last March 2006 progress report, several major design improvements have been made: - after review of the eccentric rotator design by the Department of Facilities and Planning at the Massachusetts Eye and Ear Infirmary, it was decided for safety reasons to incorporate the ceiling sliprings which were to carry all data signals in the motor base. - A larger bearing was designed to increase overall safety envelopes. - The chair was redesigned to increase subject comfort in the supine orientation used for Expts. 2A-B. Neuro Kinetics is now estimating a delivery date in the third quarter of 2007. Since proposed experiments could not be conducted, the PI conducted an alternative experiment (see details below).
1. Software and device:

a. Controller Software – Since the eccentric rotator has not been delivered as originally estimated by Neuro Kinetics, the PI decided to partially focus efforts in third year on controller and data acquisition software so that these two essential components will be ready and operational when the eccentric rotator is delivered in 2007. At the time of this progress report, both controller and data acquisition software have been developed and deployed for use with Dr. Merfeld's Swing, a device constructed by Neuro Kinetics necessitating very similar controller and data acquisition capabilities.

The controller software has been successfully migrated from an old and now technically obsolete version running under Windows 98 to a new version that runs under Windows XP. This new controller software can control both digital motors (as required for the eccentric rotator) as well as traditional analog motors used for older motion devices.

b. Data acquisition software – This new version of our data acquisition software has been developed under the Labview platform (National Instruments). This version can now simultaneously acquire up to sixteen channels and save data in various formats (binary, text) both locally and/or remotely to a data server to avoid any potential loss of data. This upgrade was necessary because the software previously used in our Lab was becoming technically obsolete since it had been designed for prior generations of National Instruments DAQ cards and it did not allow data to be saved on networked disks to ensure adequate and immediate back-up.

c. Eye movement acquisition – The Jenks Vestibular Physiology Lab has purchased a 3D-VOG Video-Oculography system (Sensomotoric Instruments, Needham, MA – USA) which is a fully integrated system for binocular acquisition and analysis of horizontal, vertical and torsional eye movements. The non-invasive Video-Oculography technique allows for comfortable recordings. Eye movements can be observed on high quality video images irrespective of whether the subject has a free-field-of-view or is in total darkness. With the help of Sensomotoric Instruments USA, we have built a customized rack-computer for our 3D-VOG Video-Oculography system. This rugged rack computer will be mounted on the eccentric rotator, and it has been designed to withstand high-G forces as well as vibrations during controlled motion.

d. Eccentric Rotator - Neuro Kinetics has ordered components and begun assembly We have been informed to expect delivery of the "Eccentric Rotator" in the third quarter of 2007. This device is being purchased primarily by this NASA grant with small contributions via other resources. The new eccentric rotator will be capable of creating three motion profiles: rotation, translation, and tilt. The main axis is a direct drive motor that rotates about an earth vertical axis. The translation axis moves the subject radially, relative to the yaw axis, at up to 2g combined acceleration. The subject will be tested in sitting and supine positions. The roll/tilt assembly will provide limited rotation (+/- 30 degrees) with the subject in either a seated position or in supine position, with head center on the tilt axis (the feet sweep an arc of 60 degrees). The system allows for independent rotation, translation (maximum radius of 70cm from center and up to 2G total head acceleration).

A Swing will be mounted on our eccentric rotator. The Swing will be able to provide tilts of up to 30° amplitude and the linear translator is capable of providing up to 2 G maximum. These motions will be combined to yield combined translation and tilt stimulation as originally proposed. As before, both motion axes (tilt and horizontal translation) will be controlled via feedback servocontrol amplifiers and custom software that is presently in use in our lab.

In the original grant application, the Eccentric Rotator was supposed to be delivered in 3 phases: tilt mechanism alone (upright orientation) in Year 1, eccentric rotator with tilt mechanism upright in Year 2, modification of tilt mechanism for supine orientation in Year 3. All designs have now been completed. Instead of delivering separately the tilting mechanism in the upright orientation and the eccentric rotator, both will be delivered simultaneously in the third quarter of 2007.

2. Alternative experiment: Influence of roll vection on interaural translation perception

Since the delivery of the Eccentric Rotator has been delayed, we decided to conduct an alternate experiment investigating tilt and translation responses similar to those originally proposed but using an existing linear track in our Lab. Ten seated restrained human subjects (5 males and 5 females, age 23 to 39) were passively translated along an earth-horizontal track aligned with their interaural axis, while constant-velocity (60°/s) roll optokinetic stimulation was applied. We utilized motion profiles similar to those previously employed [1]. Specifically, each lateral translation consisted of a single cycle of sinusoidal acceleration that had a duration of 4 seconds, which corresponded to a raised sinusoidal profile in position; the acceleration magnitude (am) was between 0 and 20 mG to the left (+) or right (-), in increments of 2 mG (1 mG ˜ 1 cm/s2); trials at 0 mG consisted of 4 seconds of random vibrations (zero-mean noise - 0.23 cm/s RMS variations) to that experienced during actual translations.

In each testing session, the direction of roll optokinetic stimulation was kept constant (clockwise or counterclockwise), and subjects randomly experienced each of the 21 lateral acceleration levels one time (10 trials to the left, 10 trials to the right, and one vibration trial with no net translation). Subjects were tested in six consecutive testing sessions, alternating clockwise and counterclockwise optokinetic stimulation with 5 min resting periods between testing sessions (the initial direction of optokinetic stimulation was counterbalanced across subjects). Instead of using a real optokinetic drum, like that used for our earlier studies [2, 3], we used virtual reality software to display a virtual optokinetic drum - with a diameter equal to the distance between two side LCD monitors (76 cm) and the end of the virtual drum located 40 cm in front of subjects at the plane of a frontal monitor. Subjects were asked to pay attention to the moving dots on the front monitor without pursuing any of them and to look "through" the center of the 8 cm diameter circular central zone that included no dots. To maximize the probability that illusory tilt was present, the operator triggered each lateral displacement when the measured subjective tilt exceeded a subject-dependent threshold for more than one second. For 7 subjects, the threshold was 10 degrees; for the remaining 3 subjects the threshold was 5 degrees due to relatively weak tilt illusions. After each lateral translation, subjects indicated their perceived direction of translation by pressing a button located on the left or right of the somatosensory tilt bar. (Since subjective roll tilt never approached 90°, "left" and

	"right" were unambiguously defined.) 3. Results: First, a single illusory tilt value - for each subject and direction of optokinetic stimulation - was obtained by averaging the subjective tilt measurements across the two-second period preceding each translation. We measured an average illusory tilt across subjects (± SE) of 10.1 ± 1.7º for CW direction and -10.2 ± 1.6º for CCW direction, which is above the threshold because the operator tried to start the motion as soon as the threshold was exceeded. These average illusory tilts are somewhat higher than previously measured (~ 7º) with a real optokinetic drum [3], which substantiates our use of virtual optokinetic stimulation. Second, data showed that the perception of translation was affected by the presence of the roll optokinetic rotation cue. Using a generalized linear model, we determined that perception of translation was significantly different between CW and CCW optokinetic stimulations ($p < 0.01$). Specifically, the normal cumulative distribution function showed that the average curve inflexion point across subjects was -1.66 mG for CW drum rotation, and +4.23 mG for CCW drum rotation. Furthermore, the threshold seems greater during optokinetic stimulation (=6.71 mG) than when measured in the dark (5.8 mG) during similar lateral translations [1]. Slope and threshold reflect similar information, and differences between dark and optokinetic conditions may be due to a decreased ability (= increased variance) to detect the direction of linear motion in the presence of the visual roll rotation cue. Therefore, we found that the reported interaural translation perception was significantly different ($p < 0.01$) during clockwise and counterclockwise optokinetic stimulation; this is consistent with the prediction that illusory tilt should be accompanied by a non-zero neural estimate of linear acceleration that contributes to translation perception. 4. Future plans: We are very grateful that NASA reinstated our funding until May 2008. Since the Eccentric Rotator won't be delivered before the third quarter of 2007, the PI proposes a revised experiment schedule - that Expt. 1 be completed by May 2008 and that Expt. 2 will be begun before the end of May 2008. Since funding ends in May 2008, a no-cost extension would then allow us to conduct the proposed experiments beyond that date. 5. References 1. Benson, A.J., M.B. Spencer, and J.R. Stott, Thresholds for the detection of the direction of whole-body, linear movement in the horizontal plane. Aviat Space Environ Med, 1986. 57(11): p. 1088-96. 2. Merfeld, D.M. and L.H. Zupan, Influence of Rotational Cues on the Neural Processing of Gravito-Inertial Force, in Levels of Perception, L.R. Harris and M. Jenkin, Editors. 2003, Springer-Verlag: New York. p. 341-373. 3. Zupan, L.H. and D.M. Merfeld, Neural processing of gravito-inertial cues in humans. IV. Influence of roll visual rotational cues on human orientation and eye movements. J Neurophysiol, 2003. 89(1): p. 390-400.
Bibliography Type:	Description: (Last Updated: 12/02/2010)
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